

The following questions are intended to guide you through the most important concepts that you must understand from the lectures and the material provided.

A good exercise would be to study the lectures/course material until you are able to answer them without having to read or search for it in the slides.

Lecture 5: Attacks (PA and MA)

To help your knowledge, we will use in the document the terminology related to attacks to biometric systems (PA, PAD, MA, etc.) You can get familiar with this terminology in slides 8 and 50

Questions: points of attack to a biometric system

- Distinction between indirect and direct attacks, where do they happen in the system and how (slides 3-6)

Questions: types of direct attacks (or presentation attacks, PA) to a biometric system

- Which are the two categories (or purposes) of PA (slide 7)
- Terminology of direct attacks: PA, PAD, impostor, concealer (slide 8)
- Which are the main methods to carry out PA (slides 19-30)

Questions: methods to counteract PA to a biometric system (or PAD methods)

- Where in a biometric system can we implement PAD (slide 33)
- What properties of a person can we measure to implement PAD (slide 34)
- How can we measure the accuracy of a PAD method (slides 35-36)
- Which specific properties of biometric modalities can we try to measure to implement PAD
 - Fingerprints (slides 37-39)
 - Iris (slides 40-42)
 - Face (slides 43-48)

Questions: morphing attacks (MA)

- How a MA is generated (slide 50) and how it can be carried out in a passport (or in principle, in any document or application needing the picture of a person, slides 52-53)
- What is the problem of a identity document having a morphed image (slides 54-56)
- Which are the two scenarios we can try to detect a MA (slide 57)